

HOYDALSMO II, Norway

WMO: 014470

Lat: 59.4969N

Lon: 8.1992E

Elev: 1844

StdP: 13.74

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-2.0	-10.7	3.3	-6.3	-5.9	4.3	-1.7	14.9	35.0	13.2	33.0	2.4	310	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	75.5	58.1	72.2	56.9	68.3	55.6	60.8	69.3	59.1	66.8	57.4	64.7	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	77.0	62.4	56.2	72.1	61.0	54.4	67.3	59.3	27.7	69.3	26.6	66.9	25.4	64.5	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	10.9	9.4	DB	-14.3	79.6	4.5	2.9	-17.6	81.7	-20.2	83.3	-22.8	84.9	-26.1	87.0	(4)
(5)			WB	-14.6	63.2	3.8	1.6	-17.4	64.4	-19.6	65.3	-21.7	66.2	-24.5	67.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	19.8	22.2	28.2	36.4	44.9	52.0	56.2	52.8	47.2	38.0	29.0	21.8		
	DBStd	14.51	9.23	9.33	8.48	6.05	7.12	5.15	5.41	4.03	4.90	5.71	7.88	10.46		
	HDD50	5039	936	779	675	409	195	34	7	16	109	373	631	875		
	HDD65	10061	1401	1199	1140	859	625	390	277	378	535	836	1081	1340		
	CDD50	459	0	0	0	0	36	95	198	103	25	2	0	0		
	CDD65	3	0	0	0	0	0	0	3	0	0	0	0	0		
	CDH74	151	0	0	0	0	8	34	108	1	0	0	0	0		
	CDH80	7	0	0	0	0	0	1	6	0	0	0	0	0		
	WSAvg	3.8	3.6	3.7	4.3	4.1	4.3	4.4	3.9	3.6	3.9	3.6	3.2	3.4		
	PrecAvg	37.4	3.4	2.0	1.9	2.1	2.6	3.1	3.8	4.2	3.6	4.1	3.6	2.9		
(15) Precipitation	PrecMax	55.9	6.9	5.3	3.8	5.0	6.7	6.1	9.3	7.2	7.1	10.8	12.0	6.1		
	PrecMin	28.0	0.4	0.4	0.2	0.6	0.2	0.4	0.3	0.3	1.7	0.8	0.8	0.3		
	PrecStd	5.9	1.6	1.3	1.0	1.2	1.5	1.5	2.0	1.9	1.5	2.3	2.2	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	45.7	55.0	62.7	74.8	78.2	80.8	72.9	68.9	58.2	48.5	45.5		
		MCWB	37.6	38.9	42.8	48.5	56.4	56.7	59.8	58.9	57.3	52.3	44.7	41.3		
	2%	DB	39.3	41.8	49.4	58.3	70.0	73.6	77.1	69.2	63.3	54.1	44.9	42.1		
		MCWB	35.9	36.9	41.1	46.2	54.0	55.0	59.4	57.9	53.6	47.6	42.2	38.6		
	5%	DB	35.7	38.6	44.3	53.7	64.9	69.6	73.8	65.7	59.8	50.8	42.1	38.9		
		MCWB	33.4	34.5	38.2	43.3	50.9	54.0	58.7	55.3	52.1	45.7	39.7	36.1		
	10%	DB	33.2	35.4	41.1	48.8	60.2	65.2	69.8	62.6	56.7	47.9	39.2	35.5		
		MCWB	31.5	32.4	35.1	40.9	49.2	52.1	57.1	54.2	50.6	43.9	37.3	33.5		
	0.4%	WB	38.4	40.4	44.8	49.8	58.5	59.2	64.1	61.1	58.6	54.7	46.2	41.7		
		MCDB	41.5	44.7	53.5	61.4	70.6	69.1	72.8	68.1	62.6	58.4	48.0	45.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.1	37.2	41.1	47.0	55.8	57.3	61.9	59.5	55.6	49.4	42.7	39.0		
		MCDB	38.6	41.1	47.8	56.5	65.8	67.7	71.6	65.3	60.1	51.7	44.6	41.7		
	5%	WB	33.8	35.0	38.7	44.2	52.8	55.7	60.3	57.8	53.8	47.0	40.1	36.2		
		MCDB	35.3	38.0	44.3	51.9	61.7	65.5	69.0	63.5	58.7	49.9	42.0	38.5		
	10%	WB	31.8	33.0	35.9	41.7	50.4	53.7	58.6	55.9	51.8	44.4	37.4	33.8		
		MCDB	33.2	35.1	40.0	48.5	57.8	62.6	66.5	61.0	55.8	47.4	39.0	35.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	17.4	19.8	19.4	20.1	19.9	18.7	17.7	16.3	14.8	12.8	13.9		
		MCDBR	15.5	19.8	23.2	28.1	27.9	28.5	27.0	23.3	22.7	19.5	15.4	14.7		
		MCWBR	13.8	15.6	15.1	16.6	13.7	12.9	11.7	12.5	13.7	13.2	12.3	12.3		
		MCDBR	15.1	17.2	22.0	25.9	23.3	23.1	20.9	18.5	18.1	15.6	13.3	14.3		
	5% WB	MCWBR	13.3	13.6	14.5	15.9	12.4	11.8	10.6	11.4	12.0	11.9	11.5	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.230	0.267	0.297	0.336	0.338	0.329	0.347	0.341	0.313	0.297	0.249	0.205		
	taud		2.194	2.278	2.268	2.288	2.432	2.493	2.453	2.460	2.537	2.498	2.287	2.102		
	Ebn at Noon		189	241	267	275	283	287	279	272	263	231	180	152		
	Edh at Noon		13	21	29	34	32	31	31	29	23	17	12	10		
(44) All-Sky Solar Radiation	RadAvg		106	314	777	1279	1521	1715	1507	1191	774	367	138	70		
	RadStd		22	69	95	202	213	161	201	94	102	69	34	8		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page